

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Final
 Duration: **2 Hours**

Semester: **Spring 2026**
 Full Marks: **70**

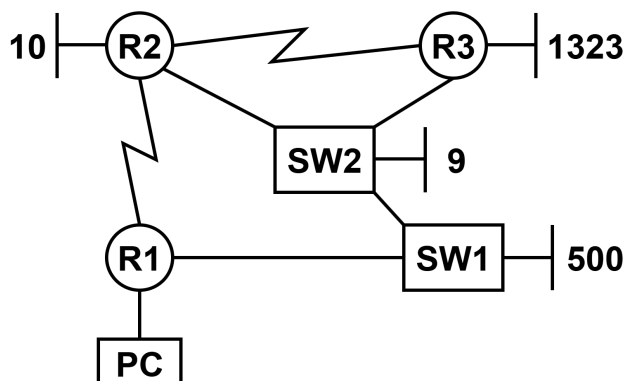
CSE421 / EEE465 : Computer Networks

Answer **Sections A, B, and C** as per the instructions given. (**Pages: 4**)

Figures in the right margin indicate marks.

Name:	ID:	Section:
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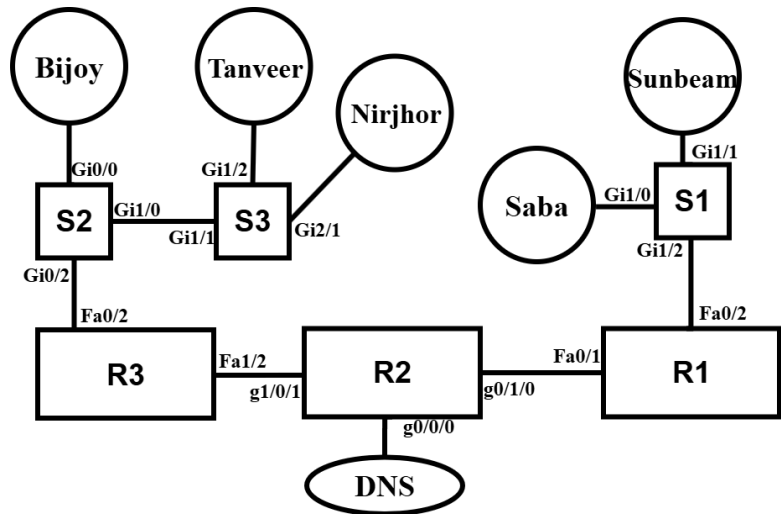
SECTION A [All questions of this section are MANDATORY] - 40 MARKS

Q1 [CO3]	Consider the network topology shown in the figure on the right side, where the numbers of hosts specified for each network represent only the end devices. I. A university has been given the IP address 103.84.77.219/20. First, find the network address. II. Using variable-length subnet masking (VLSM), allocate appropriate IP address ranges to each network from the network address that you calculated from Q1(I). You have to show the hierarchical tree.		3 + 9
Q2	A router receives an IPv4 datagram with a total size of 7,340 bytes, including a 20-byte IP header. The datagram must be forwarded through a network with an outgoing link MTU of 1,300 bytes. After fragmentation, one of the fragments is captured. The captured fragment has the following values: Total Length = 1,300 bytes More Fragments (MF) = 1 Fragment Offset = 320	3 + 2 + 2 + 3 + 3	
[CO3]	I. Which fragment was captured? Number the first fragment as Fragment 1.		
[CO3]	II. How many fragments will be created from the original datagram in total?		
[CO3]	III. What will be the Total Length field value of the last fragment?		
[CO3]	IV. What will be the Fragment Offset value of the last fragment?		
[CO2]	V. What information does the destination host need in order to correctly reassemble fragmented IPv4 packets?		

Q3
[CO2]

Routers R1, R2, and R3 use PAT. R1 acts as a DHCP server for its LAN and uses private IP address **10.10.10.1** and public IP address **212.5.5.5**. It assigns host IP addresses starting from **10.10.10.2**.

R3 also acts as a DHCP server for its own LAN and uses private IP address **10.10.10.1** and public IP address **212.9.9.9**. It assigns host IP addresses starting from **10.10.10.2**.



All hosts use the same DNS resolver, whose IP address is **6.6.6.6**. All switches start with empty MAC tables with a 2-minute aging time, and all ARP caches are initially empty. Each host's MAC address is its name (e.g., SUNBEAM).

- I. Apart from its own IP address, what other information does Saba receive from DHCP?
- II. If Sunbeam and Saba both send DHCPDISCOVER messages to R1, how does R1 distinguish between the two requests? Explain.
- III. For Nirjhor to send a packet to Saba for the first time, assuming all ARP caches are initially empty. How many ARP requests are required in total? Mention each ARP request.
- IV. Nirjhor and Saba both use the private IP address 10.10.10.2 in their respective LANs and send packets to the DNS server using source port 45000. Can their requests still be uniquely identified by the DNS server? Explain.
- V. After 2 minutes of inactivity, all switch MAC table entries expire. Tanveer then sends an ICMP Echo Request to Bijoy, and later Bijoy sends an ARP request for Nirjhor. Which of these transmitted frames will cause the switches to learn MAC addresses again? Explain.

END OF SECTION A

[CO3] **SECTION B** [Answer ANY TWO out of THREE in this section] - **12 MARKS**

Q4

- I. A network administrator runs Traceroute to a destination and notices that the first few hops respond normally, but later hops show "Request Timed Out." Does this always mean the destination is unreachable? Explain.
- II. Give two possible reasons for this behavior.

Q5

- I. Write the expanded hexadecimal form of the following IPv6 addresses:
A. fdc2:34::ab:0
B. ::1

3
+
3
+
3
+
3
+
3

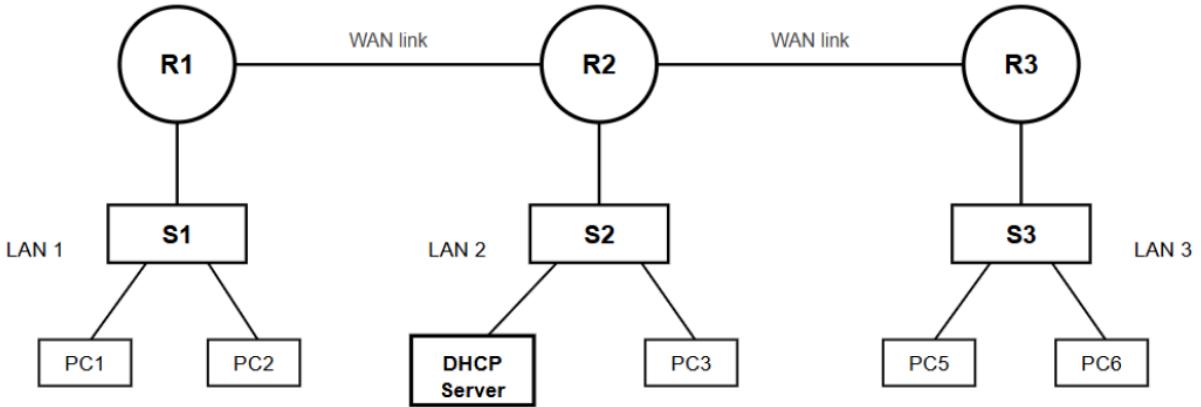
3
+
3

2
+
4

	II. Name the type of each address and state whether each address is routable on the Internet.	
Q6	If only Switch 2 currently has Bijoy's MAC address in its MAC table, use the topology given in Q3 to answer the following questions. I. Tanveer wants to send a unicast data frame to Bijoy. Identify which devices will receive the frame and what each device will do with it. II. Draw the MAC tables of all switches after Bijoy sends the reply.	3 + 3

END OF SECTION B

[CO2] SECTION C [Answer ANY THREE out of FIVE in this section] - 18 MARKS

Q7	I. Beyond simply counting hops, what is the primary purpose of the Time-to-Live (TTL) field, and what would happen to network bandwidth if this field were omitted? II. Why does the IPv4 header include a Header Checksum field even though some other layers also perform error detection?	3 + 3
Q8	I. Explain why Link State protocols require continuous monitoring of neighbour link status, while Distance Vector protocols rely mainly on routing information periodically received from neighbouring routers. II. Explain why Link State routing is generally considered more stable and adaptive than Distance Vector routing in rapidly changing networks.	3 + 3
Q9	 The DHCP server is located in LAN 2, connected through switch S2 under router R2. A DHCP relay agent is configured on R1. However, PCs 5 and 6 still cannot obtain IP addresses. A technician claims: "Since R1 already has a DHCP relay configured, LAN 3 requests will automatically reach the DHCP server." I. Is this claim correct? Justify your answer. II. 50% of PC1's DHCP lease time has elapsed. What DHCP messages will PC1 exchange with the DHCP server at this point?	3 + 3
Q10	I. Both Anycast and Multicast involve multiple possible receivers. Explain how the delivery logic of Anycast differs from that of Multicast. II. Imagine two IPv6-only networks separated by a large IPv4-only transit network.	3 + 3

	Explain why Dual Stack alone may not be sufficient for communication between them, and what kind of additional mechanism might be needed.	
Q11	I. A network printer is replaced with a new model that keeps the same IP address but has a different physical NIC. Explain why a PC on the same network might be unable to print to it for several minutes, even though the IP configuration is correct. II. Switches are often described as transparent devices. Using a path of three interconnected switches as an example, explain why the sender's link-layer header remains unchanged as the frame passes through these devices.	3 + 3

END OF SECTION C

===== **THE END** =====

I'd tell you a joke about UDP, but you might not get it.